

BIO 005 · HUMAN PHYSIOLOGY · WEEK 2

Competency list.

The Cell · Transport · Signaling

Name _____

Date _____

These are the twenty five things you have to be able to do by Sunday. Every exam question, lab and case in Week 2 comes from this list. Tick a box only when it is true.

HOW TO USE THE THREE BOXES

Read. You have read it, in the book or the notes, and it made sense at the time.

Drew it. You have drawn or worked it with the notes open beside you.

From memory. You did it on blank paper with nothing open. This is the only box that predicts an exam score, and it is the one to chase. A row with the first two ticked and the third empty is exactly where your study time should go.

Cell anatomy and tissues **STRUCTURE**

4 competencies

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|----------|--|----------------------------------|--|--|
| 1 | Body fluid compartments
State the approximate volumes of total body water, intracellular fluid, extracellular fluid, plasma, and interstitial fluid in a 70 kg adult and compare the dominant solutes of the intracellular and extracellular compartments. | ☐
READ | ☐
DREW
IT | ☐
FROM
MEMORY |
| <hr/> | | | | |
| 2 | Compartment separation and clinical volume shifts
Predict the direction of water movement between compartments when extracellular osmolarity rises or falls and name a clinical situation that produces each shift. | ☐
READ | ☐
DREW
IT | ☐
FROM
MEMORY |
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| 3 | Membrane composition and fluidity
Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties. | ☐
READ | ☐
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FROM
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| 4 | Determinants of permeability
Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein. | ☐
READ | ☐
DREW
IT | ☐
FROM
MEMORY |

Cell physiology and transport **WHAT CROSSES THE MEMBRANE**

12 competencies

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|----------|--|----------------------------------|--|--|
| 5 | Simple diffusion and Fick's law
State the variables in Fick's law of diffusion and predict how a change in concentration gradient, surface area, membrane thickness, or distance changes the rate of transfer. | ☐
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| 6 | Osmolarity and tonicity
Calculate the osmolarity of a solution, distinguish osmolarity from tonicity, and classify a solution as isotonic, hypotonic, or hypertonic to a cell. | ☐
READ | ☐
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| 7 | Osmosis and cell volume
Predict the direction of water movement and the resulting change in cell volume when a cell is placed in a solution of stated osmolarity and penetrating solute content. | ☐
READ | ☐
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FROM
MEMORY |

8	Diffusion and osmosis experiment Measure diffusion and osmotic movement across a selectively permeable membrane and relate the observed rate to molecular size and concentration gradient.	☐ READ	☐ DREW IT	☐ FROM MEMORY
9	Tonicity and red blood cells Observe erythrocytes in solutions of different tonicity, identify crenation, normal shape, and hemolysis, and explain each result.	☐ READ	☐ DREW IT	☐ FROM MEMORY
10	Facilitated diffusion Describe carrier and channel mediated diffusion and explain why carrier mediated transport shows saturation and specificity while simple diffusion does not.	☐ READ	☐ DREW IT	☐ FROM MEMORY
11	Primary active transport Explain how the sodium potassium ATPase uses ATP to move three sodium out and two potassium in, and state the two gradients it maintains.	☐ READ	☐ DREW IT	☐ FROM MEMORY
12	Secondary active transport Distinguish symport from antiport and trace how the sodium gradient powers glucose uptake by SGLT and calcium removal by the sodium calcium exchanger.	☐ READ	☐ DREW IT	☐ FROM MEMORY
13	Transport maximum and saturation Interpret a transport rate curve, identify the transport maximum, and apply the concept to renal glucose handling in hyperglycemia.	☐ READ	☐ DREW IT	☐ FROM MEMORY
14	Vesicular transport Compare phagocytosis, pinocytosis, receptor mediated endocytosis, and exocytosis by trigger, cargo, and energy requirement.	☐ READ	☐ DREW IT	☐ FROM MEMORY
15	Transepithelial transport Trace glucose or sodium from lumen to blood across a polarized epithelium and identify which step occurs at the apical membrane and which at the basolateral membrane.	☐ READ	☐ DREW IT	☐ FROM MEMORY
16	Transport simulation Use a membrane transport simulation to distinguish simple diffusion from facilitated diffusion and active transport by their response to gradient reversal and metabolic poison.	☐ READ	☐ DREW IT	☐ FROM MEMORY

Cell signaling HOW CELLS TALK TO EACH OTHER

9 competencies

17	Signal types and range Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.	☐ READ	☐ DREW IT	☐ FROM MEMORY
18	Receptor location and ligand solubility Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.	☐ READ	☐ DREW IT	☐ FROM MEMORY

19 G protein coupled receptors

Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

☐

READ

☐DREW
IT☐FROM
MEMORY**20 Second messengers**

Identify cyclic AMP, IP₃, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

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READ

☐DREW
IT☐FROM
MEMORY**21 Catalytic receptors and intracellular receptors**

Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

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READ

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MEMORY**22 Signal amplification**

Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

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READ

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IT☐FROM
MEMORY**23 Receptor modulation**

Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

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READ

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MEMORY**24 Signal termination**

Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

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READ

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MEMORY**25 Dose response relationships**

Plot a dose response curve, identify threshold, maximal response, and EC₅₀, and compare a full agonist with a partial agonist on the same axes.

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READ

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IT☐FROM
MEMORY